

NEET Important Chapter – Electrostatic Potential and Capacitance

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NEET

In this chapter we understand how we move the charge in this field and what amount of work is done to move the charge against this field and in what form the work done is stored in the system and what is electrostatic potential.

In this chapter we first define electrostatic potential and then potential due to a point charge, system of charges, electric dipole and potential energy of charges and dipole in an external field. We got introduced to new concepts like equipotential surface, electrostatics of conductors, dielectric and polarization etc. and define electrostatic potential energy.

In this chapter students also get to learn about the capacitors and capacitance and the effect of dielectric medium on it and their combination circuits and electrostatic potential and capacitance all formulas.

Now, let us move on to the important concepts and formulae related to [NEET 2022](#) exams along with a few solved examples.

Important Topics of Electrostatic Potential and Capacitance

- Electrostatic Potential and Capacitance Notes
- Equipotential Surface
- Capacitor and Capacitance
- Energy Stored in a Capacitor
- Series Combination of Capacitors
- Electrostatic Potential
- Capacitors in the Parallel Combination
- Parallel Plate Capacitor Derivation
- Potential Energy of System of Charges

Important Concepts of Electrostatic Potential and Capacitance

Name of the Concept	Key Points of the Concept
1. Electrostatic potential	• The amount of work done to bring a unit positive charge from infinite to a given position in the electric field is known as electric potential. $V = W/q$ • Electric potential due to a point charge q $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ here r is the distance from the charge to the point where the potential is to be found. S.I. unit - Joule/Coulomb or volt

2. Electric potential difference	• The amount of work done to bring a unit positive charge from one position to the other position in the electric field is known as Electric potential difference. • S.I. unit - Joule/Coulomb or volt
3. Electrostatic potential energy	• The electric potential energy is the required energy to move a charge against in electric field • Potential energy of a single charge - $U = qV$ Here V is the potential • Potential energy of a system of two charges- $U(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$ here r is the distance between the charges S.I. unit - Joule

4. Electrostatic with conductor	• Electrostatic field is zero inside a conductor because there is no charge inside the conductor. • All charges are at the surface of the conductor and at every point on the surface of the charged conductor there is normal electric field only. • Electric potential is constant or the same value inside the conductor and surface of the conductor. • Electrostatic shielding - there is no influence of outside electric charge and field in the cavity inside the conductor. The electric field inside the conductor is always zero so any cavity in conductor is shielded from outside and it is known as electrostatic shielding
5. Polarisation	• The dipole moment per unit volume is called polarisation and denoted by P
6. Capacitance	• It is the ratio of charges given to the conductor and its potential. • Capacity of conductor = $C = \text{Charge} / \text{potential}$ • Unit - farad • It depends on shape and size of conductor and surrounding and presence of other charge objects near the conductor.

8. Parallel plate capacitor	• It consists of two metal plates parallel to each other separated by a medium. • Used to accumulate electric charge or increase the electrical capacity. • Capacitance of capacitor is increased when it is placed in dielectric medium.
7. Combination of capacitors	• Capacitor in series - In series combination, charges are same on the plate of each capacitor and total potential drop is sum of individual potential drop of capacitors. • Capacitor in parallel - In parallel combination, the same potential difference is applied across all capacitors that are connected and charges are different.
9. Van de graaff generator	• It is a device which is used for accelerating the charged particles.

List of Important Formulae

Sl. No	Name of the Concept	Formula
1.	Work done by external force in moving a charge q from Q to P	
2.	Potential due to a point charge	_____ here _____

3.	Potential due to an electric dipole	
4.	Potential due to a system of charges	
5.	Relation between electric field and potential	
6.	Potential energy of system of charges (3 charge)	
7.	Potential energy of a dipole in an external field	
8.	Capacitance of parallel plate capacitor	
9.	Equivalent Capacitance in series combination	
10.	Equivalent Capacitance in parallel combination	

Solved Examples

1. A parallel plate capacitor with air between the plates has a capacitance of 8pF ($1\text{pF} = 10^{-12}\text{F}$). What will be the capacitance if the distance between the plates is reduced by half and the space between them is filled with a substance of dielectric constant 6?

Sol:

Given, Capacitance, $C = 8\text{pF} =$ _____ F

As per the question In the first case, the parallel plates are at a distance 'd' and

We know that air has dielectric constant, $k = 1$ so capacitance in first case

Capacitance - _____

Here A = area of the plates

ϵ_0 = permittivity of free space

Now, according to the question if the distance between the plates is halved then

$$= d/2$$

Given, dielectric constant of the substance, _____ = 6

Hence, the capacitance of the capacitor-

By solving this we get

$$C_1 = 96 \text{ PF}$$

Key point: If a parallel plate capacitor is placed in any medium with dielectric constant k then its capacitance is increased by k times.

2. A regular hexagon of the side 10 cm has a charge $5 \mu\text{C}$ at each of its vertices. Calculate the potential at the center of the hexagon.

Sol: The given figure shows a regular hexagon with charges at each vertex.

Here, given - $q = 5 \mu\text{C} =$

$AB = BC = CD = DE = EF = FA = \text{side of hexagon} = 10 \text{ cm}.$

By the diagram the distance of the vertices from the center O, $r = 10 \text{ cm}.$

The electric potential at point O,

Here,

$\epsilon_0 =$ Permittivity of free space and

_____ $\times \quad \times \quad \times$

Trick: If we have symmetry in the shape then we just determine the potential due to a single charge and then multiply it with the total number of charges there are in the system.

Previous Year Questions from NEET Paper

1. The equivalent capacitance of the combination shown in the figure is- (NEET 2021)

- (a) $3C$
- (b) $2C$
- (c) $C/2$
- (d) $3C/2$

Sol: In the above figure the capacitor in the right side mesh is not contribute in the equivalent capacitance because there is same potential across the both terminal of the capacitor so no current flow through it therefore only capacitors which is in the left mesh is considered and they are in parallel combination so equivalent capacitance would be

$$C_{eq} = C + C = 2C$$

Option (b) is the correct option.

Key point - Current is flow due to potential difference.

2. In a certain region of space with volume 0.2

, the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is (NEET 2021)

(a) zero

(b) 0.5 N/C

(c) 1 N/C

(d) 5 N/C

Sol: We know the relation between potential and electric field

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In the given question potential is found to be 5 V throughout the volume (region) so change in potential is zero therefore electric field is also zero in this region.

Option (a) is the correct option.

Trick – Electric field is equal to the potential gradient with negative sign.

Practice Questions

1. If potential (in volts) in a region is expressed as $V(x,y,z) = 6x - y + 2yz$, the electric field (in N/C) at point $(1,1,0)$ is ? (**Ans:** -)
2. Two capacitors with capacity C_1 and C_2 , when connected in series, have capacitance C_s and when connected in parallel have capacitance C_p . find the relation between capacitances ? (**Ans:** $C_p = 4C_s$)

Conclusion

In this article we have provided important information regarding the chapter electrostatic potential formula and capacitance such as important concepts, formulae, etc.. Students should work on more solved examples for securing good grades in the NEET exams.

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